

CS 2601 Linear and Convex Optimization

10. KKT conditions

Bo Jiang

John Hopcroft Center for Computer Science
Shanghai Jiao Tong University

Fall 2023

Active and inactive constraints

Let $\mathbf{x} \in \mathbb{R}^n$ and $n \geq k$. Consider

$$\begin{array}{ll}\min_{\mathbf{x}} & f(\mathbf{x}) \\ \text{s.t.} & h_i(\mathbf{x}) = 0, \quad i = 1, 2, \dots, k \\ & g_j(\mathbf{x}) \leq 0, \quad j = 1, 2, \dots, m\end{array} \quad (\text{ICP})$$

We do **not** assume it is a convex problem. Assume the domain is $\mathbb{R}^n$.
The feasible set is

$$X = \{\mathbf{x} : h_i(\mathbf{x}) = 0, \quad 1 \leq i \leq k; \quad g_j(\mathbf{x}) \leq 0, \quad 1 \leq j \leq m\}$$

Let $\mathbf{x}_0 \in X$. The j -th inequality constraint $g_j(\mathbf{x}) \leq 0$ is called **active** at $\mathbf{x}_0$ if $g_j(\mathbf{x}_0) = 0$, and **inactive** at $\mathbf{x}_0$ if $g_j(\mathbf{x}_0) < 0$. Denote by $J(\mathbf{x}_0)$ the set of the indices of the active inequality constraints at $\mathbf{x}_0$,

$$J(\mathbf{x}_0) = \{j : g_j(\mathbf{x}_0) = 0\}$$

By convention, equality constraints are considered active at all $\mathbf{x} \in X$. 1

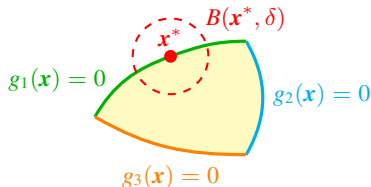
Reduction to equality constrained problem

Suppose $\mathbf{x}^*$ is a local minimum of (ICP),
i.e. $f(\mathbf{x}) \geq f(\mathbf{x}^*)$ for all $\mathbf{x}$ such that

$$h_i(\mathbf{x}) = 0, \quad i = 1, 2, \dots, k$$

$$g_j(\mathbf{x}) \leq 0, \quad j = 1, 2, \dots, m$$

$$\mathbf{x} \in B(\mathbf{x}^*, \delta)$$



Since $g_j(\mathbf{x}^*) < 0$ for $j \notin J(\mathbf{x}^*)$, for small δ ,

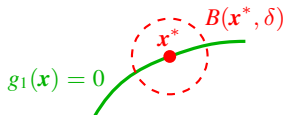
$$\mathbf{x} \in B(\mathbf{x}^*, \delta) \implies g_j(\mathbf{x}) < 0, \quad j \notin J(\mathbf{x}^*)$$

Thus $f(\mathbf{x}) \geq f(\mathbf{x}^*)$ for all $\mathbf{x}$ such that

$$h_i(\mathbf{x}) = 0, \quad i = 1, 2, \dots, k$$

$$g_j(\mathbf{x}) = 0, \quad j \in J(\mathbf{x}^*)$$

$$\mathbf{x} \in B(\mathbf{x}^*, \delta)$$



Reduction to equality constrained problem (cont'd)

A local minimum $\mathbf{x}^*$ of (ICP) is also a local minimum of the following

$$\begin{array}{ll}\min_{\mathbf{x}} & f(\mathbf{x}) \\ \text{s.t.} & h_i(\mathbf{x}) = 0, \quad i = 1, 2, \dots, k \\ & g_j(\mathbf{x}) = 0, \quad j \in J(\mathbf{x}^*)\end{array}$$

$\mathbf{x}^* \in X$ is a **regular point** if $\nabla h_i(\mathbf{x}^*)$, $1 \leq i \leq k$ and $\nabla g_j(\mathbf{x}^*)$, $j \in J(\mathbf{x}^*)$ are linearly independent.

At a regular local minimum, Lagrange condition yields

$$\nabla f(\mathbf{x}^*) + \sum_{i=1}^k \lambda_i^* \nabla h_i(\mathbf{x}^*) + \sum_{j \in J(\mathbf{x}^*)} \mu_j^* \nabla g_j(\mathbf{x}^*) = \mathbf{0}$$

Setting $\mu_j^* = 0$ for inactive constraints, i.e. $j \notin J(\mathbf{x}^*)$,

$$\nabla f(\mathbf{x}^*) + \sum_{i=1}^k \lambda_i^* \nabla h_i(\mathbf{x}^*) + \sum_{j=1}^m \mu_j^* \nabla g_j(\mathbf{x}^*) = \mathbf{0}$$

Karush-Kuhn-Tucker (KKT) conditions

Theorem. If $\mathbf{x}^*$ is a local minimum of (ICP) and also a regular point, then there exist **Lagrange multipliers**¹ $\lambda_1^*, \dots, \lambda_k^*, \mu_1^*, \dots, \mu_m^* \in \mathbb{R}$ s.t. the following KKT conditions hold,

1. $\nabla f(\mathbf{x}^*) + \sum_{i=1}^k \lambda_i^* \nabla h_i(\mathbf{x}^*) + \sum_{j=1}^m \mu_j^* \nabla g_j(\mathbf{x}^*) = \mathbf{0}$ (**stationarity**)
2. $\mu_j^* \geq 0, j = 1, 2, \dots, m$
3. $\mu_j^* g_j(\mathbf{x}^*) = 0, j = 1, 2, \dots, m$ (**complementary slackness**)

Note. Condition 1 says $\nabla_x \mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) = \mathbf{0}$ for the **Lagrangian**

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = f(\mathbf{x}) + \boldsymbol{\lambda}^T \mathbf{h}(\mathbf{x}) + \boldsymbol{\mu}^T \mathbf{g}(\mathbf{x}) = f(\mathbf{x}) + \sum_{i=1}^k \lambda_i h_i(\mathbf{x}) + \sum_{j=1}^m \mu_j g_j(\mathbf{x})$$

Note. Condition 3 is called **complementary slackness condition**, as it implies either $\mu_j^* = 0$ or $g_j(\mathbf{x}^*) = 0$.

¹Sometimes also called KKT multipliers. Sometimes λ_i are called Lagrange multipliers while μ_j are called KKT multipliers.

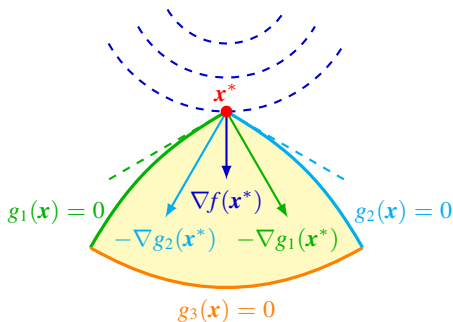
Geometric interpretation

Let $\mathbf{x} \in \mathbb{R}^2$. Consider

$$\begin{array}{ll}\min_{\mathbf{x}} & f(\mathbf{x}) \\ \text{s.t.} & g_j(\mathbf{x}) \leq 0, j = 1, 2, 3\end{array}$$

Suppose $\mathbf{x}^*$ is a local minimum and only g_1 and g_2 are active at $\mathbf{x}^*$. The KKT condition says $\mu_1^* \geq 0$, $\mu_2^* \geq 0$, $\mu_3^* = 0$ and

$$\nabla f(\mathbf{x}^*) = -\mu_1^* \nabla g_1(\mathbf{x}^*) - \mu_2^* \nabla g_2(\mathbf{x}^*)$$



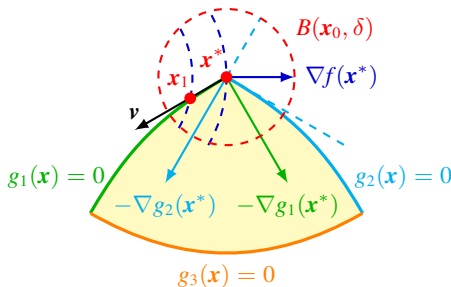
Geometric interpretation (cont'd)

Why $\mu_j^* \geq 0$? Assume $\mu_2^* < 0$ and we show a contradiction.

- Let $\mathbf{v} = -\nabla g_2(\mathbf{x}^*) - \mathcal{P}_{-\nabla g_1(\mathbf{x}^*)}(-\nabla g_2(\mathbf{x}^*))$
- $-\nabla g_2(\mathbf{x}^*) \notin \text{span}\{-\nabla g_1(\mathbf{x}^*)\}$ by the regularity of $\mathbf{x}^*$, so $\mathbf{v} \neq \mathbf{0}$.
- $\mathbf{v}^T \nabla g_1(\mathbf{x}^*) = 0$, $\mathbf{v}^T \nabla g_2(\mathbf{x}^*) = -\|\mathbf{v}\|^2 < 0$.
- Move along $g(x) = 0$ in the direction of $\mathbf{v}$ from $\mathbf{x}^*$ to e.g. $\mathbf{x}_1$. Since $\mu_2^* < 0$,

$$\mathbf{v}^T \nabla f(\mathbf{x}^*) = \mathbf{v}^T [-\mu_1^* \nabla g_1(\mathbf{x}^*) - \mu_2^* \nabla g_2(\mathbf{x}^*)] = -\mu_2^* \mathbf{v}^T \nabla g_2(\mathbf{x}^*) < 0,$$

For a small step, $f(\mathbf{x}_1) < f(\mathbf{x}^*)$, contradicting minimality of $f(\mathbf{x}^*)$.



Proof of $\mu \geq 0$

Suppose $\mu_{j_0}^* < 0$ for some j_0 . We will show we can move away from $\mathbf{x}^*$ so that feasibility is maintained but f decreases, contradicting the minimality of $\mathbf{x}^*$. Let $J'(\mathbf{x}^*) = J(\mathbf{x}^*) \setminus \{j_0\}$, and

$$S = \text{span} \{ \nabla h_i(\mathbf{x}), i = 1, 2, \dots, k; \nabla g_j(\mathbf{x}), j \in J'(\mathbf{x}^*) \}$$

1. $\nabla g_{j_0}(\mathbf{x}^*) \notin S$ by regularity of $\mathbf{x}^*$.
2. Let $\mathbf{v} = -\nabla g_{j_0}(\mathbf{x}^*) - \mathcal{P}_S(-\nabla g_{j_0}(\mathbf{x}^*))$. Then $\mathbf{v} \perp S$, $\mathbf{v}^T \nabla g_{j_0}(\mathbf{x}^*) < 0$
3. KKT then implies $\mathbf{v}^T \nabla f(\mathbf{x}^*) < 0$
4. There exists a path $\mathbf{x}(t)$ s.t. $\mathbf{x}(0) = \mathbf{x}^*$, $\mathbf{x}'(0) = \mathbf{v}$, $h_i(\mathbf{x}(t)) = 0, \forall i$, and $g_j(\mathbf{x}(t)) = 0$ for $j \in J'(\mathbf{x}^*)$.
5. For $j \notin J(\mathbf{x}^*)$, $g_j(\mathbf{x}^*) < 0$. By continuity, $g_j(\mathbf{x}(t)) < 0$ for small t .
6. By the chain rule,

$$\left. \frac{d}{dt} g_{j_0}(\mathbf{x}(t)) \right|_{t=0} = \nabla g_{j_0}(\mathbf{x}^*)^T \mathbf{x}'(0) = \nabla g_{j_0}(\mathbf{x}^*)^T \mathbf{v} < 0$$

For small $t > 0$, $g_{j_0}(\mathbf{x}(t)) < g_{j_0}(\mathbf{x}^*) = 0$. Similarly, $f(\mathbf{x}(t)) < f(\mathbf{x}^*)$.

Proof of $\mu \geq \mathbf{0}$ (cont'd)

Proof of step 3.

- By KKT Conditions 2 and 3

$$\nabla f(\mathbf{x}^*) = - \sum_{i=1}^k \lambda_i^* \nabla h_i(\mathbf{x}^*) - \sum_{j \in J(\mathbf{x}^*)} \mu_j^* \nabla g_j(\mathbf{x}^*) - \underbrace{\sum_{j \notin J(\mathbf{x}^*)} \mu_j^* \nabla g_j(\mathbf{x}^*)}_{=0}$$

- Since $\mathbf{d} \perp S$, $\mu_{j_0}^* < 0$,

$$\begin{aligned} \mathbf{d}^T \nabla f(\mathbf{x}^*) &= - \sum_{i=1}^k \lambda_i^* \underbrace{\mathbf{d}^T \nabla h_i(\mathbf{x}^*)}_{=0} - \sum_{j \in J'(\mathbf{x}^*)} \mu_j^* \underbrace{\mathbf{d}^T \nabla g_j(\mathbf{x}^*)}_{=0} - \mu_{j_0}^* \underbrace{\mathbf{d}^T \nabla g_{j_0}(\mathbf{x}^*)}_{<0} \\ &< 0 \end{aligned}$$

Alternative approach

Assume no equality constraints.

Consider a path $\mathbf{x}(t) : [0, \epsilon) \rightarrow X$ s.t. $\mathbf{x}(0) = \mathbf{x}^*$ and $\mathbf{v} = \mathbf{x}'(0_+)$.

For $j \in J(\mathbf{x}^*)$,

$$g_j(\mathbf{x}(t)) \leq 0 = g_j(\mathbf{x}^*) \implies \mathbf{v}^T \nabla g_j(\mathbf{x}^*) = \lim_{t \downarrow 0} \frac{g_j(\mathbf{x}(t)) - g_j(\mathbf{x}(0))}{t} \leq 0$$

Regularity guarantees the set of feasible directions at $\mathbf{x}^*$ is precisely

$$F_{\mathbf{x}^*}X = \{\mathbf{v} : \mathbf{v}^T \nabla g_j(\mathbf{x}^*) \leq 0, j \in J(\mathbf{x}^*)\}.$$

At a local minimum $\mathbf{x}^*$,

$$\left. \frac{d}{dt} f(\mathbf{x}(t)) \right|_{t=0_+} = \mathbf{v}^T \nabla f(\mathbf{x}^*) \geq 0$$

This should be true for every $\mathbf{v} \in F_{\mathbf{x}^*}X$.

Alternative approach (cont'd)

Normal cone of $F_{\mathbf{x}^*}X$ at $\mathbf{0}$,

$$N_{\mathbf{x}^*}X = \{\mathbf{n} : \mathbf{n}^T \mathbf{v} \leq 0, \forall \mathbf{v} \in F_{\mathbf{x}^*}X\},$$

The optimality condition means

$$-\nabla f(\mathbf{x}^*) \in N_{\mathbf{x}^*}X$$

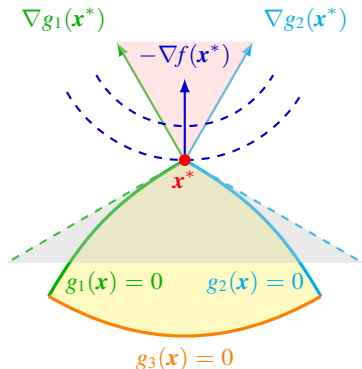
It turns out

$$\begin{aligned} N_{\mathbf{x}^*}X &= \text{cone}\{\nabla g_j(\mathbf{x}^*), j \in J(\mathbf{x}^*)\} \\ &= \left\{ \sum_{j \in J(\mathbf{x}^*)} \mu_j \nabla g_j(\mathbf{x}^*) : \mu_j \geq 0 \right\} \end{aligned}$$

so

$$-\nabla f(\mathbf{x}^*) = \sum_{j \in J(\mathbf{x}^*)} \mu_j^* \nabla g_j(\mathbf{x}^*)$$

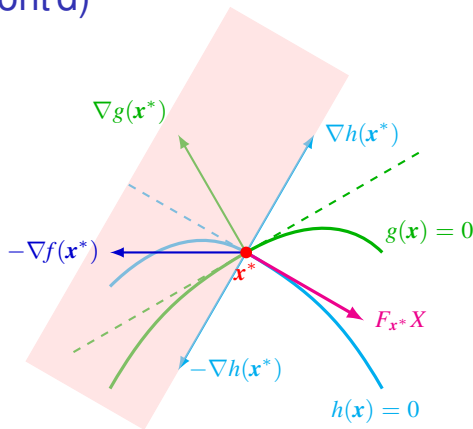
for some $\mu_j^* \geq 0$.



Alternative approach (cont'd)

Consider

$$\begin{array}{ll}\min_{\mathbf{x}} & f(\mathbf{x}) \\ \text{s.t.} & h(\mathbf{x}) = 0 \\ & g(\mathbf{x}) \leq 0\end{array}$$



Replace $h(\mathbf{x}) = 0$ by $h(\mathbf{x}) \leq 0$ and $-h(\mathbf{x}) \leq 0$. By KKT conditions,

$$\begin{aligned}-\nabla f(\mathbf{x}^*) &= \mu^* \nabla g(\mathbf{x}^*) + \tilde{\mu}^* \nabla h(\mathbf{x}^*) + \hat{\mu}^* (-\nabla h(\mathbf{x}^*)) \\ &= \mu^* \nabla g(\mathbf{x}^*) + \lambda^* \nabla h(\mathbf{x}^*)\end{aligned}$$

where $\lambda^* = \tilde{\mu}^* - \hat{\mu}^* \in \mathbb{R}$.

Sufficiency of KKT conditions for convex problems

Theorem. For a convex problem, i.e. f and g_j are convex, and h_i are affine, if there exist $\lambda_1^*, \dots, \lambda_k^*$ and $\mu_1^*, \dots, \mu_m^*$ s.t. the KKT conditions are satisfied at a feasible $\mathbf{x}^* \in X$, then $\mathbf{x}^*$ is a global minimum.

Note. The previous necessary conditions assume $\mathbf{x}^*$ is regular point. The sufficient conditions here assume convexity but **not** regularity.

Proof. It suffices to show $\nabla f(\mathbf{x}^*)^T(\mathbf{x} - \mathbf{x}^*) \geq 0, \forall \mathbf{x} \in X$.

1. By the KKT conditions,

$$\nabla f(\mathbf{x}^*)^T(\mathbf{x} - \mathbf{x}^*) = - \sum_i \lambda_i^* \nabla h_i(\mathbf{x}^*)^T(\mathbf{x} - \mathbf{x}^*) - \sum_{j \in J(\mathbf{x}^*)} \mu_j^* \nabla g_j(\mathbf{x}^*)^T(\mathbf{x} - \mathbf{x}^*)$$

It suffices to show $\nabla h_i(\mathbf{x}^*)^T(\mathbf{x} - \mathbf{x}^*) = 0$ and $\nabla g_j(\mathbf{x}^*)^T(\mathbf{x} - \mathbf{x}^*) \leq 0$.

2. Since $h_i(\mathbf{x}) = \mathbf{a}_i^T \mathbf{x} - b_i$ is affine, and $h_i(\mathbf{x}) = h(\mathbf{x}^*) = 0$ by feasibility,

$$\nabla h_i(\mathbf{x}^*)^T(\mathbf{x} - \mathbf{x}^*) = \mathbf{a}_i^T(\mathbf{x} - \mathbf{x}^*) = h_i(\mathbf{x}) - h(\mathbf{x}^*) = 0$$

3. For $j \in J(\mathbf{x}^*)$, $g_j(\mathbf{x}^*) = 0$ and $g_j(\mathbf{x}) \leq 0$. By the convexity of g_j ,

$$\nabla g_j(\mathbf{x}^*)^T(\mathbf{x} - \mathbf{x}^*) \leq g_j(\mathbf{x}) - g_j(\mathbf{x}^*) \leq 0$$

Example

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^3} \quad & f(\mathbf{x}) = x_1 + 2x_2 + x_3 \\ \text{s.t.} \quad & h(\mathbf{x}) = x_1 + x_2 + 2x_3 = 0 \\ & g(\mathbf{x}) = \|\mathbf{x}\|^2 - 1 \leq 0 \end{aligned}$$

All feasible points are regular. The Lagrangian is

$$\mathcal{L}(\mathbf{x}, \lambda, \mu) = x_1 + 2x_2 + x_3 + \lambda(x_1 + x_2 + 2x_3) + \mu(x_1^2 + x_2^2 + x_3^2 - 1)$$

The KKT conditions (including the constraints) are

$$\begin{cases} \mu \geq 0 \\ \partial_{x_1} \mathcal{L} = 1 + \lambda + 2\mu x_1 = 0 \\ \partial_{x_2} \mathcal{L} = 2 + \lambda + 2\mu x_2 = 0 \\ \partial_{x_3} \mathcal{L} = 1 + 2\lambda + 2\mu x_3 = 0 \\ \mu(x_1^2 + x_2^2 + x_3^2 - 1) = 0 \\ x_1 + x_2 + 2x_3 = 0 \\ x_1^2 + x_2^2 + x_3^2 - 1 \leq 0 \end{cases}$$

Example (cont'd)

Case I. g is inactive. Thus $\mu = 0$. But this leads to a contradiction.

$$\begin{cases} \partial_{x_1} \mathcal{L} = 1 + \lambda + 2\mu x_1 = 0 \implies \lambda = -1 \\ \partial_{x_2} \mathcal{L} = 2 + \lambda + 2\mu x_2 = 0 \implies \lambda = -2 \end{cases}$$

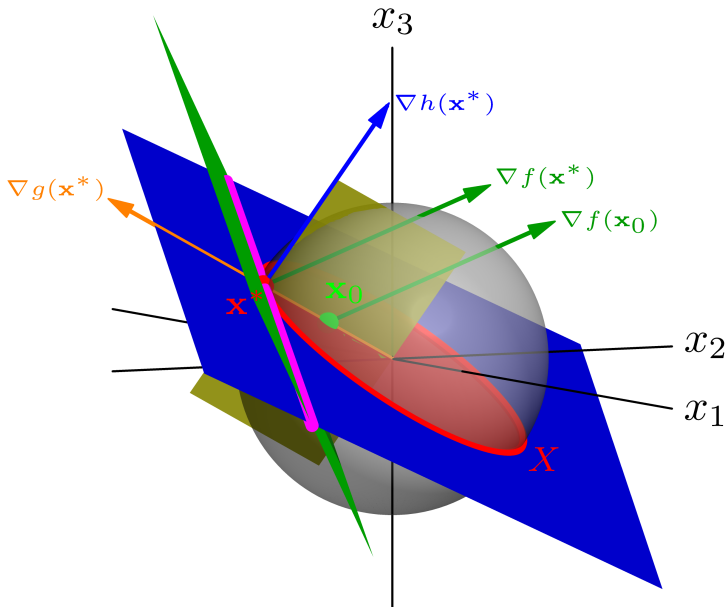
Case II. g is active. This essentially reduces to an example for Lagrange condition, but we only take the solution with $\mu \geq 0$,

$$\begin{cases} x_1 = -\frac{1}{\sqrt{66}} \\ x_2 = -\frac{7}{\sqrt{66}} \\ x_3 = \frac{4}{\sqrt{66}} \\ \lambda = -\frac{5}{6} \\ \mu = \sqrt{\frac{33}{72}} \end{cases}$$

Since the problem is convex, the above gives a global minimum.

Note. By minimizing $-f$, one can verify the other solution for the example for Lagrange condition is a global maximum.

Example (cont'd)



Example

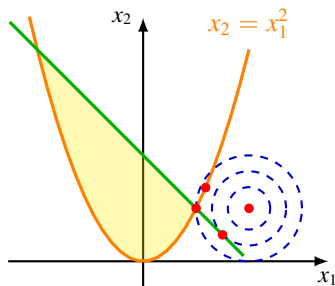
$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^2} \quad & f(\mathbf{x}) = (x_1 - 2)^2 + (x_2 - 1)^2 \\ \text{s.t.} \quad & g_1(\mathbf{x}) = x_1^2 - x_2 \leq 0 \\ & g_2(\mathbf{x}) = x_1 + x_2 - 2 \leq 0 \end{aligned}$$

All feasible points are regular. The Lagrangian is

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\mu}) = (x_1 - 2)^2 + (x_2 - 1)^2 + \mu_1(x_1^2 - x_2) + \mu_2(x_1 + x_2 - 2)$$

The KKT conditions (including the constraints) are

$$\begin{cases} \mu_1 \geq 0 \\ \mu_2 \geq 0 \\ \partial_{x_1} \mathcal{L} = 2(x_1 - 2) + 2\mu_1 x_1 + \mu_2 = 0 \\ \partial_{x_2} \mathcal{L} = 2(x_2 - 1) - \mu_1 + \mu_2 = 0 \\ \mu_1(x_1^2 - x_2) = 0 \\ \mu_2(x_1 + x_2 - 2) = 0 \\ x_1^2 - x_2 \leq 0 \\ x_1 + x_2 - 2 \leq 0 \end{cases}$$



Example (cont'd)

Case I. Both g_1 and g_2 are inactive, so $\mu_1 = \mu_2 = 0$.

$$\begin{cases} \partial_{x_1} \mathcal{L} = 2(x_1 - 2) = 0 \\ \partial_{x_2} \mathcal{L} = 2(x_2 - 1) = 0 \end{cases} \implies \begin{cases} x_1 = 2 \\ x_2 = 1 \end{cases}$$

But

$$x_1^2 - x_2 = 3 > 0, \quad x_1 + x_2 - 2 = 1 > 0$$

violating $g_1 \leq 0$ and $g_2 \leq 0$.

Case II. g_2 is active, but g_1 is inactive, so $\mu_1 = 0$.

$$\begin{cases} \partial_{x_1} \mathcal{L} = 2(x_1 - 2) + \mu_2 = 0 \\ \partial_{x_2} \mathcal{L} = 2(x_2 - 1) + \mu_2 = 0 \\ x_1 + x_2 - 2 = 0 \end{cases} \implies \begin{cases} x_1 = \frac{3}{2} \\ x_2 = \frac{1}{2} \\ \mu_2 = 1 \end{cases}$$

But

$$x_1^2 - x_2 = \frac{7}{4} > 0$$

violating $g_1 \leq 0$.

Example (cont'd)

Case III. g_1 is active, but g_2 is inactive, so $\mu_2 = 0$.

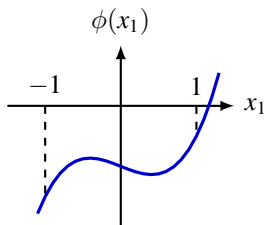
$$\begin{cases} \partial_{x_1} \mathcal{L} = 2(x_1 - 2) + 2\mu_1 x_1 = 0 \\ \partial_{x_2} \mathcal{L} = 2(x_2 - 1) - \mu_1 = 0 \\ x_1^2 - x_2 = 0 \end{cases}$$

From the last two equations,

$$x_2 = x_1^2, \quad \mu_1 = 2(x_2 - 1) = 2(x_1^2 - 1)$$

Plugging into the first equation,

$$2(x_1 - 2) + 4x_1(x_1^2 - 1) = 0 \implies \phi(x_1) \triangleq 2x_1^3 - x_1 - 2 = 0$$



Note $\mu_1 \geq 0 \implies x_2 \geq 1 \implies x_1^2 \geq 1 \implies x_1 \geq 1 \text{ or } x_1 \leq -1$.

If $x_1 \geq 1$, then $x_1 + x_2 \geq 2$, contradicting $x_1 + x_2 < 2$ (g_2 is inactive).

If $x_1 \leq -1$, $\phi(x_1) = 0$ has no solution since $\phi'(x_1) = 6x_1^2 - 1 > 0$ for $x_1 \leq -1$ and $\phi(-1) = -3 < 0$.

Example (cont'd)

Case IV. Both g_1 and g_2 are active.

$$\begin{cases} x_1^2 - x_2 = 0 \\ x_1 + x_2 - 2 = 0 \end{cases} \implies \begin{cases} x_1 = 1 \\ x_2 = 1 \end{cases} \quad \text{or} \quad \begin{cases} x_1 = -2 \\ x_2 = 4 \end{cases}$$

Plugging into

$$\begin{cases} \partial_{x_1} \mathcal{L} = 2(x_1 - 2) + 2\mu_1 x_1 + \mu_2 = 0 \\ \partial_{x_2} \mathcal{L} = 2(x_2 - 1) - \mu_1 + \mu_2 = 0 \end{cases}$$

yields

$$\begin{cases} x_1 = 1 \\ x_2 = 1 \\ \mu_1 = \frac{2}{3} \\ \mu_2 = \frac{2}{3} \end{cases} \quad \text{or} \quad \begin{cases} x_1 = -2 \\ x_2 = 4 \\ \mu_1 = -\frac{14}{3} \quad (\text{violating } \mu_1 \geq 0) \\ \mu_2 = -\frac{32}{3} \quad (\text{violating } \mu_2 \geq 0) \end{cases}$$

This is a convex problem, so $\mathbf{x}^* = (1, 1)$ is the global minimum.

Example: Power allocation

Consider the power allocation problem,

$$\begin{aligned} \max_{P_1, \dots, P_n} \quad & \sum_{i=1}^n W_i \log_2 \left(1 + \frac{P_i}{N_i} \right) \\ \text{s.t.} \quad & \sum_{i=1}^n P_i \leq P \\ & P_i \geq 0, \quad i = 1, 2, \dots, n \end{aligned}$$

The optimal solution should satisfy $\sum_{i=1}^n P_i = P$ (why?). Equivalent to

$$\begin{aligned} \min_{\mathbf{P}} \quad & f(\mathbf{P}) = - \sum_{i=1}^n W_i \log \left(1 + \frac{P_i}{N_i} \right) \\ \text{s.t.} \quad & h(\mathbf{P}) = \sum_{i=1}^n P_i - P = 0 \\ & g_i(\mathbf{P}) = -P_i \leq 0, \quad i = 1, 2, \dots, n \end{aligned}$$

Example: Power allocation (cont'd)

The Lagrangian is

$$\mathcal{L}(\mathbf{P}, \lambda, \boldsymbol{\mu}) = - \sum_{i=1}^n W_i \log\left(1 + \frac{P_i}{N_i}\right) + \lambda \left(\sum_{i=1}^n P_i - P\right) - \sum_{i=1}^n \mu_i P_i$$

All feasible points are regular. By the stationarity condition,

$$\partial_{P_i} \mathcal{L} = -\frac{W_i}{P_i + N_i} + \lambda - \mu_i = 0, \quad i = 1, 2, \dots, n.$$

For each i ,

1. If g_i is inactive, then $\mu_i = 0$ and

$$P_i = \frac{W_i}{\lambda} - N_i > 0$$

2. if g_i is active, then $P_i = 0$ and

$$-\frac{W_i}{N_i} + \lambda = \mu_i \geq 0 \implies \frac{W_i}{\lambda} - N_i \leq 0$$

Example: Power allocation (cont'd)

In both cases,

$$P_i = \left(\frac{W_i}{\lambda} - N_i \right)^+ = W_i \left(r - \frac{N_i}{W_i} \right)^+, \quad \text{where } r = \frac{1}{\lambda}$$

It remains to solve for r from

$$\sum_{i=1}^n P_i = \sum_{i=1}^n W_i \left(r - \frac{N_i}{W_i} \right)^+ = P$$

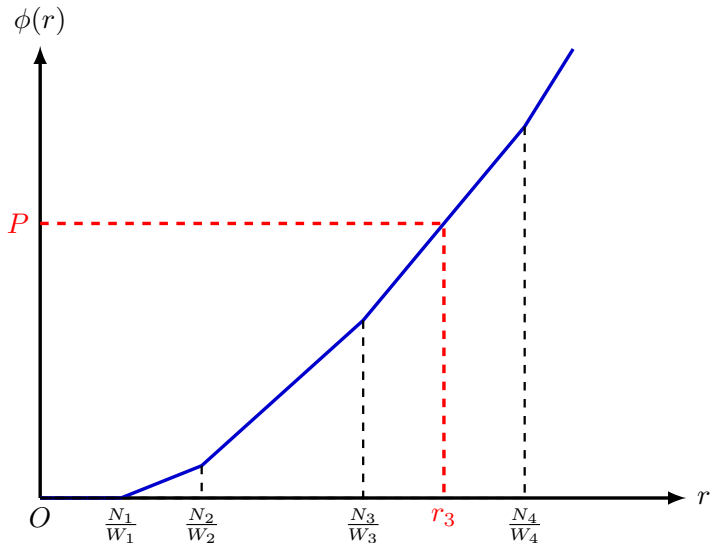
WLOG, assume $\frac{N_1}{W_1} \leq \frac{N_2}{W_2} \leq \dots \leq \frac{N_n}{W_n}$. Note $\phi(r) = \sum_{i=1}^n W_i \left(r - \frac{N_i}{W_i} \right)^+$ is continuous, piecewise linear, strictly increasing on $[\frac{N_1}{W_1}, +\infty)$ with $\phi(\frac{N_1}{W_1}) = 0$. Thus $\phi(r) = P$ has a unique solution $r = r_{k_0}$, where

$$r_k = \frac{P + \sum_{i=1}^k N_i}{\sum_{i=1}^k W_i}$$

and

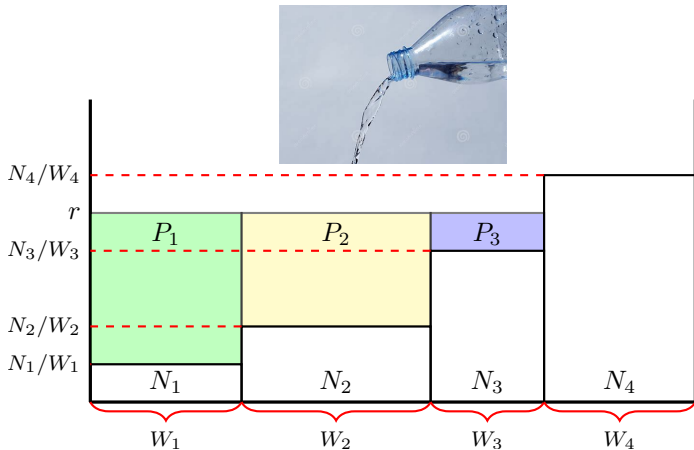
$$k_0 = \max \left\{ k : \phi\left(\frac{N_k}{W_k}\right) \leq P \right\} = \max \left\{ k : r_k \geq \frac{N_k}{W_k} \right\}$$

Example: Power allocation (cont'd)



Example: Power allocation (cont'd)

Famous **water filling** solution.



Example: SVM

Recall the hard-margin SVM,

$$\begin{aligned} \min_{\mathbf{w}, b} \quad & \frac{1}{2} \|\mathbf{w}\|^2 \\ \text{s. t.} \quad & y_i(\mathbf{w}^T \mathbf{x}_i + b) \geq 1, \quad i = 1, 2, \dots, m \end{aligned}$$

The Lagrangian is

$$\mathcal{L}(\mathbf{w}, b, \mu) = \frac{1}{2} \|\mathbf{w}\|^2 + \sum_{i=1}^m \mu_i [1 - y_i(\mathbf{w}^T \mathbf{x}_i + b)]$$

By the stationarity condition

$$\nabla_{\mathbf{w}} \mathcal{L} = \mathbf{w}^* - \sum_{i=1}^m \mu_i^* y_i \mathbf{x}_i = \mathbf{0} \implies \mathbf{w}^* = \sum_{i=1}^m \mu_i^* y_i \mathbf{x}_i$$

By the complementary slackness condition, $\mu_i^* = 0$ if $y_i(\mathbf{w}^T \mathbf{x}_i + b) > 1$, i.e. if $\mathbf{x}_i$ does not lie on the margin boundary.

Example: SVM (cont'd)

The points x_i on the margin boundary are called **support vectors**.

w^* is a linear combination of support vectors.

$$w^* = \sum_{i: x_i \text{ is support vector}} \mu_i^* y_i x_i$$

